

Project Problem Statement and Task-wise Division

Indian-Context-Aware Multilingual and Multimodal Web Retrieval & Summarization

Project objective

Build an end-to-end system that accepts a query in Hindi/Devanagari, Roman Hindi, or English; dynamically retrieves relevant web information; extracts text from webpages, PDFs, scans and images; prioritizes evidence that is especially relevant to the Indian context; and produces a concise multi-document answer together with an interpretable reliability assessment.

Important principle: Indian-context relevance and source reliability are different. A source should not be trusted merely because it is Indian, and an international source should not be ignored merely because it is foreign. For Indian cultural or historical questions, India-relevant primary, textual, scholarly, institutional and local evidence should receive stronger contextual priority, while reliability must be assessed independently.

Illustrative Example

User query (Devanagari): “राम कैसे थे? राम को क्यों भगवान मानते हैं लोग?”

Equivalent Roman Hindi: “Ram kaise the, Ram ko kyu Bhagwan mante hain log?”

For this question, useful evidence may appear in very different forms:

Possible source	What it may contain	How the system should treat it
English academic / encyclopedic page	An English source may explain Rama from literary, historical or comparative perspectives.	Useful external perspective; not automatically dominant.
Hindi / Devanagari webpage	A source may discuss Indian cultural, philosophical or religious understandings in Hindi.	High contextual relevance when directly responsive to the query.
Scanned book or manuscript page	The relevant text may exist only as an image or scanned page, for example a Sanskrit/Hindi passage in Devanagari.	Requires OCR before retrieval and summarization.
Temple / museum / archival photograph	A photograph may contain Devanagari text on a board, inscription, caption or exhibit.	OCR confidence must be considered before using the extracted text.
Indian scholarly / institutional source	A university, archive, research article or institution may provide historical or textual explanation.	Can provide both strong contextual relevance and stronger evidential support.

Key idea for students

The task is not “find Indian websites and summarize them.” The task is to retrieve heterogeneous evidence, understand which evidence is most relevant to the user’s Indian-context question, keep reliable external evidence for corroboration, and synthesize the answer without treating all sources equally.

Overall System Workflow

User query (Hindi / Devanagari / Roman Hindi / English)
↓
Query understanding + multilingual search queries
↓
On-the-fly web search and permitted content acquisition
↓
HTML / PDF / image / scanned-document detection
↓
Direct text extraction OR English/Devanagari OCR
↓
Text cleaning, normalization and de-duplication
↓
Passage retrieval and Indian-context relevance ranking
↓
Evidence selection + source/support signals
↓
Multi-document summarization (Open-MDS as a starting framework)
↓
Final answer + supporting sources + reliability assessment

Task-wise Division (5 Tasks)

Task	Module	Primary responsibility
Task 1	Multilingual Query Understanding & Search Query Generation	Convert Hindi/Devanagari/Roman-Hindi input into effective Hindi and English search queries.
Task 2	On-the-Fly Web Search, Scraping & Content Acquisition	Find relevant URLs and collect accessible HTML, PDFs and images while respecting access/licensing constraints.
Task 3	English + Devanagari OCR & Text Preparation	Extract machine-readable text from scans/images and normalize direct/OCR text into a common format.
Task 4	Indian-Context Relevance & Evidence Prioritization	Rank passages using query relevance, Indian-context relevance and basic source/evidence signals.

Task	Module	Primary responsibility
Task 5	Multi-Document Summarization, Reliability & Integration	Adapt Open-MDS-style summarization to dynamically collected evidence and return answer + sources + confidence.

Task 1: Multilingual Query Understanding & Search Query Generation

Problem to solve: Given a user question in Devanagari, Roman Hindi or English, generate multiple search formulations that can retrieve useful Indian and international evidence without changing the original information need.

Main activities

- Detect whether the input is Devanagari/Hindi, Roman Hindi, English, or mixed.
- Normalize spelling variants and common Roman-Hindi forms.
- Generate equivalent or complementary search queries in Devanagari/Hindi and English.
- Identify important named entities, concepts and contextual terms from the user question.
- Avoid query expansion that changes the meaning or introduces a preferred answer.
- Evaluate whether multilingual query expansion improves retrieval of useful evidence.

Input → Output

Input: User question, e.g., “राम कैसे थे? राम को क्यों भगवान मानते हैं लोग?”

Output: A ranked set of Hindi/Devanagari and English search queries + extracted key concepts

Research question

How can a short, informal Hindi or Roman-Hindi question be expanded into multilingual search queries while preserving the user’s intended meaning?

Expected deliverable

A callable module/API such as `generate_queries(user_query) → {language, normalized_query, search_queries, key_terms}`.

Task 2: On-the-Fly Web Search, Scraping & Multimodal Content Acquisition

Problem to solve: Dynamically find relevant sources for the generated queries and collect usable webpages, PDFs and images at query time rather than relying on a pre-collected dataset.

Main activities

- Retrieve top-k search results for the multilingual search queries and merge duplicate URLs.
- Study which sources can be accessed/processed under appropriate licences or terms (e.g., CC0, CC BY, CC BY-SA, open government/institutional material where permitted).
- Respect robots.txt, website terms, rate limits and access restrictions.
- Collect HTML pages, linked PDFs and relevant images when permitted.
- Store metadata: URL, title, author/organization, publication date (if available), domain, language and content type.

- Identify content that should be sent to direct extraction versus OCR.

Input → Output

Input: Search queries from Task 1

Output: Raw document package containing accessible HTML/PDF/image content + source metadata

Research question

How can heterogeneous web evidence be collected on the fly for a user query while remaining traceable and respecting access/reuse constraints?

Expected deliverable

A callable module/API such as `search_and_collect(search_queries, top_k)` → `raw_documents`.

Task 3: English + Devanagari OCR & Text Preparation

Problem to solve: Convert webpages, text PDFs, scanned PDFs and images into clean machine-readable text, with special emphasis on English and Devanagari/Hindi/Sanskrit material encountered during live retrieval.

Main activities

- Automatically detect whether a PDF already contains usable text or requires OCR.
- Evaluate freely available transformer-based OCR options for English and Devanagari; keep a practical non-transformer OCR baseline for comparison if useful.
- Process printed Devanagari, scanned Hindi/Sanskrit books, English scans, mixed-script pages and text embedded in photographs.
- Preserve page/image provenance so extracted text can be traced back to its source.
- Record OCR confidence when available and flag low-confidence text.
- Normalize Unicode, remove obvious OCR artefacts, repair line/paragraph breaks and prepare clean passages.
- Perform basic exact/near-duplicate text removal so copied pages are not over-counted.

Input → Output

Input: Raw HTML/PDF/image documents from Task 2

Output: Clean text passages + language/script + extraction method + OCR confidence + source/page reference

Research question

How accurately and efficiently can open OCR systems extract English and Devanagari text from heterogeneous web documents in an on-the-fly setting?

Expected deliverable

A callable module/API such as `extract_and_prepare(document)` → passages with provenance and OCR confidence.

Task 4: Indian-Context Relevance & Evidence Prioritization

Problem to solve: Given many extracted passages, determine which ones are most useful for answering the particular user question, with stronger contextual priority for evidence that directly reflects the relevant Indian textual, cultural, historical, institutional or local context.

Main activities

- Compute query–passage relevance so unrelated text is removed.
- Define what “Indian-context relevance” means beyond merely using an .in domain or an Indian publisher.
- Build a baseline using India-related entities/terms, Devanagari/Indic textual cues, places, institutions, primary texts and contextual vocabulary.
- Build a semantic baseline using multilingual sentence embeddings.
- Explore a learned relevance model or contrastive-learning approach after the baseline is stable.
- Keep contextual relevance separate from source reliability. An India-relevant source may still be weak evidence; a reliable international source may still be highly useful.
- Include simple source/evidence features initially (source type, attribution, date, citations, corroboration, duplication) without requiring a full learned reliability model in the first prototype.

Input → Output

Input: User query + clean passages + source metadata from Tasks 1–3

Output: Ranked evidence passages with query-relevance, Indian-context-relevance and basic source/evidence signals

Research question

How can the system prioritize evidence that is genuinely useful for an Indian-context question without equating “Indian source” with “reliable source”?

Expected deliverable

A callable module/API such as `rank_evidence(query, passages)` → scored/ranked passages.

Task 5: Multi-Document Summarization, Reliability Assessment & End-to-End Integration

Problem to solve: Generate one concise, source-grounded answer from the top-ranked evidence and communicate how strongly the final answer is supported. AllenAI Open-MDS should be studied as a starting framework for retrieval-based multi-document summarization.

Main activities

- Study the AllenAI Open-MDS repository: <https://github.com/allenai/open-mds/tree/main>
- Understand its document indexing, retrieval and summarization pipeline, and identify what must change for a temporary query-specific document collection created on the fly.
- Select diverse, non-redundant evidence before summarization so one repeated source does not dominate.
- Generate the final answer in the user’s preferred language (initially Hindi/Devanagari may be prioritized for Hindi questions).

- Attach or list the supporting sources used for major claims wherever feasible.
- Estimate answer reliability using interpretable factors such as source quality, evidence agreement, context relevance, OCR confidence, evidence coverage and unresolved contradictions.
- For the first prototype, prefer labels such as High / Medium / Low with an explanation rather than an unjustified percentage such as “91% reliable”.
- Integrate the interfaces of Tasks 1–4 into one end-to-end demonstrator and define common evaluation tests.

Input → Output

Input: User query + ranked evidence passages + source/OCR/context signals

Output: Final multi-document answer + supporting sources + interpretable reliability assessment

Research question

Can an Open-MDS-style framework be adapted to dynamically retrieved multilingual evidence and produce an Indian-context-aware answer whose support and uncertainty are transparent?

Expected deliverable

A working prototype such as `answer = summarize_with_reliability(query, ranked_evidence)`, plus a simple demo interface/API.

Illustrative Evidence Ranking for the Rama Query

The following scores are hypothetical and only demonstrate the intended logic; they are not factual evaluations of any real website.

ID	Source type	Query rel.	India rel.	Source signal	OCR conf.	Possible use
S1	English academic source	0.88	0.62	High	—	Use for scholarly/external perspective
S2	Indian scholarly Hindi source	0.94	0.95	High	—	Strong priority
S3	Scanned Sanskrit/Hindi book page	0.86	0.97	Medium/High	0.93	Use if provenance and OCR are adequate
S4	Temple photograph with Devanagari board	0.58	0.88	Medium	0.89	Supporting context only
S5	Anonymous high-overlap blog	0.90	0.96	Low	—	Do not let high Indian relevance override weak reliability

Illustrative Final Output Format

Example answer structure (illustrative only)

उत्तर: भारतीय परंपराओं में राम की व्याख्या धार्मिक, साहित्यिक और सांस्कृतिक दृष्टियों से की जाती है। अंतिम उत्तर को उन स्रोतों से प्रमुख समर्थन लेना चाहिए जो प्रश्न के भारतीय संदर्भ को सीधे समझाते हैं, जबकि विश्वसनीय अकादमिक या बाहरी स्रोतों को तुलनात्मक/पुष्टिकरण साक्ष्य के रूप में भी रखा जा सकता है। यदि अलग-अलग परंपराओं या स्रोतों में भिन्न व्याख्याएँ हों, तो सारांश को उस भिन्नता को स्पष्ट रूप से बताना चाहिए।

Reliability: HIGH (illustrative)

Why: multiple independent sources support the main points; India-context evidence is strong; OCR-extracted text used in the answer has high confidence; no major unresolved contradiction was found.

Sources used: S1, S2, S3, ...

Common Evaluation Expectations

Module	Suggested evaluation
Task 1 – Query generation	Human relevance of generated queries; retrieval gain over the original query; preservation of intent.
Task 2 – Web retrieval	Precision@K / useful-source rate; coverage of HTML/PDF/image types; duplicate URL rate; traceability.
Task 3 – OCR	Character Error Rate (CER), Word Error Rate (WER), script-wise performance, OCR confidence calibration.
Task 4 – Ranking	Precision/nDCG/Recall for relevant passages; agreement with human Indian-context relevance labels; ablation of relevance features.
Task 5 – Summarization	Coverage, relevance, redundancy, readability, factual/source support, citation correctness, and human judgment of reliability explanation.

What Each Task Team Must Submit Initially

Each task team should have a minimum of 2 and a maximum of 3 students. Each team must prepare a one-page write-up containing:

1. The exact sub-problem the team will address.
2. Input and expected output of the module.
3. Proposed workflow / methodology.
4. Existing open-source models, tools, papers or GitHub repositories to be studied.
5. Data or test cases needed to evaluate the module.
6. Evaluation metrics and a simple baseline.
7. Expected first-phase deliverable and how it will connect to the next task.